

Stern-Gerlach Experiment

Luca Cicu - Lorenzo Liuzzo

27/11/2025

Historical Context

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- 1. Hello everyone, and welcome. My name is [Your Name], and today I'm going to talk about one of the most beautiful and pivotal experiments in the history of physics: the Stern-Gerlach experiment.*
- 2. Conducted in 1922, this experiment was designed to test a specific prediction of the 'old' quantum theory. But what it ended up doing was far more profound: it accidentally discovered a fundamental, intrinsic property of matter that has no classical counterpart at all. It revealed the existence of spin.*

Hello world

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Experimental Apparatus

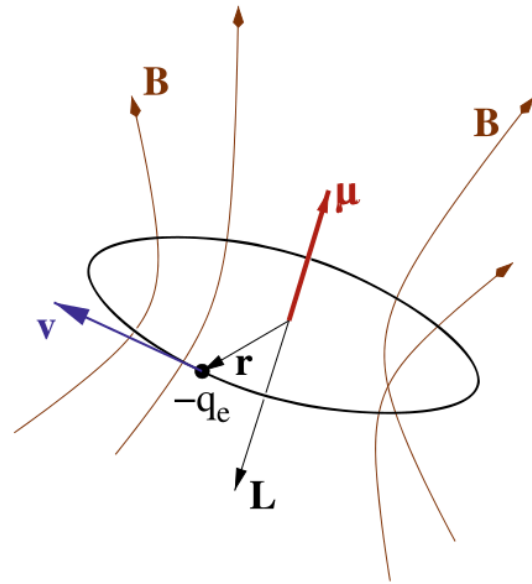
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Stern-Gerlach Experiment

Angular & Magnetic momenta

Figure 1: Angular and magnetic momenta generated by an electron orbiting circularly.



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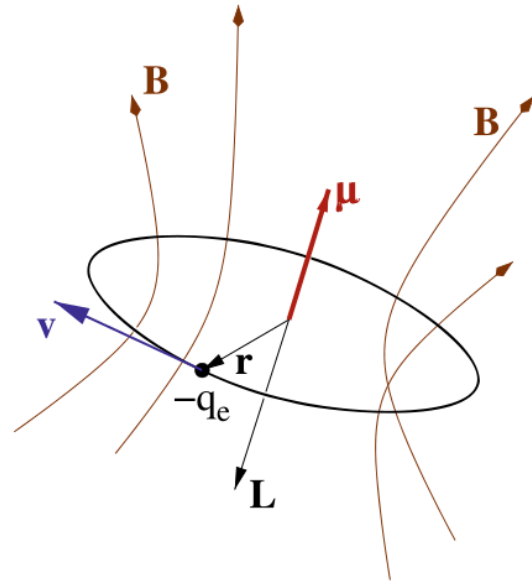
– Angular & Magnetic momenta

1. (Click) As you can see from the derivation, this magnetic moment μ is directly proportional to the angular momentum L . The proportionality constant is $q/(2m)$, the charge-to-mass ratio of the electron.
 1. This is the crucial link: if you have angular momentum, you have a magnetic moment. By measuring the magnetic moment, you are indirectly measuring the angular momentum.
2. (Click) And how do we probe a magnetic moment? With an external magnetic field, B . The interaction energy of a dipole in a field is given by this Hamiltonian: $H = -\boldsymbol{\mu} \cdot \mathbf{B}$. The dipole will want to align with the field to minimize this energy, just like a compass needle.

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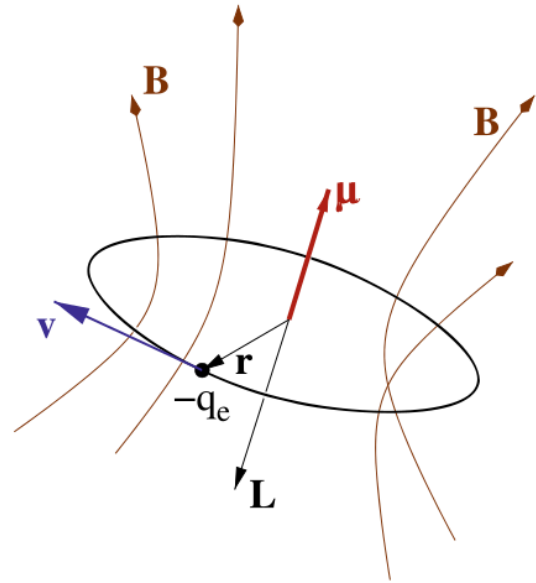
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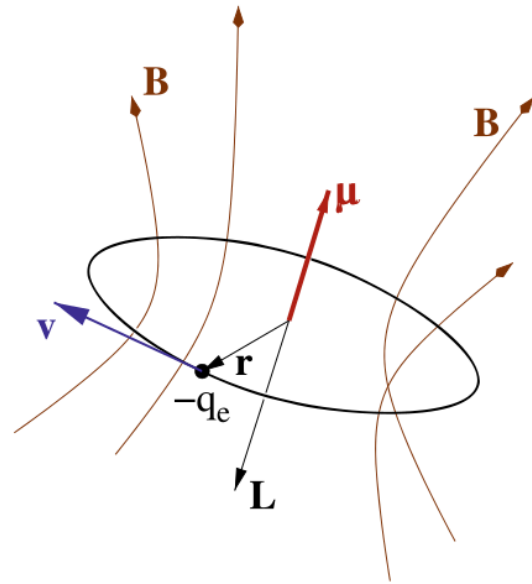
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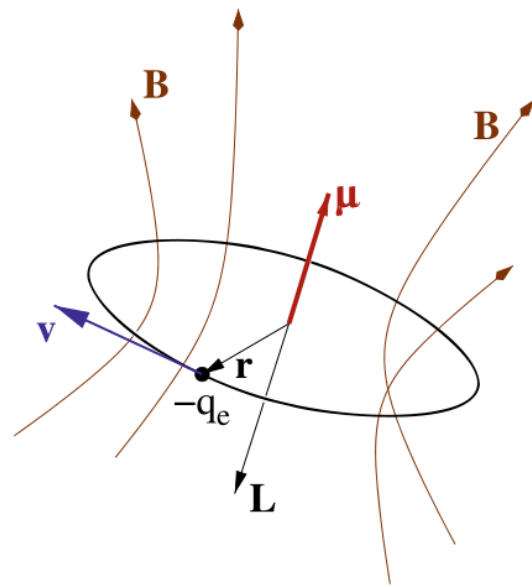
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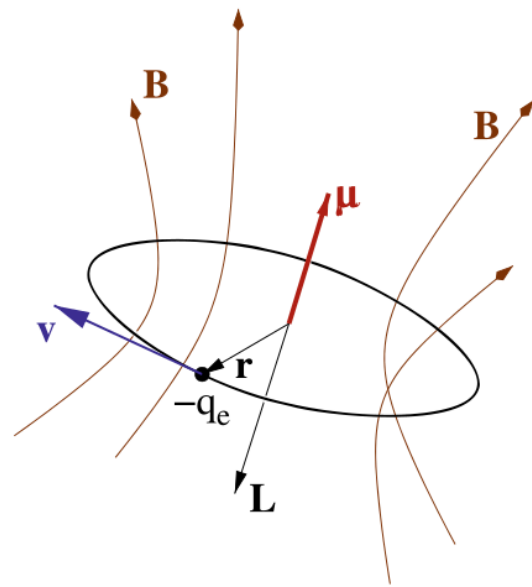
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The **interaction** between an atomic magnetic momentum and an **external magnetic field** is represented by the magnetic hamitonian

$$H_{\text{magn}} = -\boldsymbol{\mu} \cdot \mathbf{B}$$

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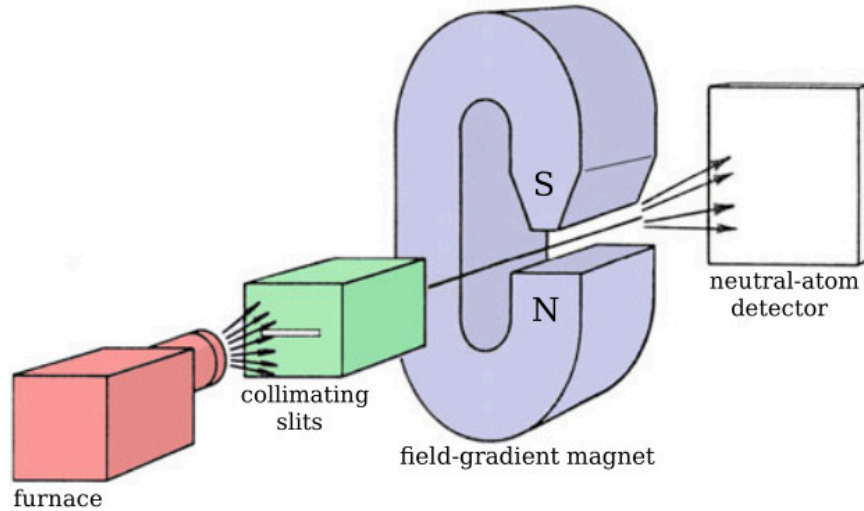
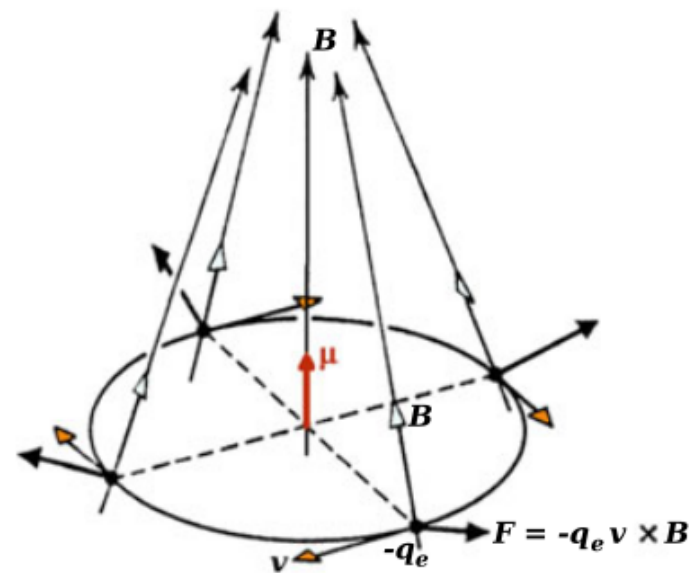


Figure 2: Stern-Gerlach apparatus.



$$F_{\text{magn}} = -\nabla H_{\text{magn}} = \nabla(\mu \cdot B)$$

Figure 3: Microscopic origin of the magnetic force.

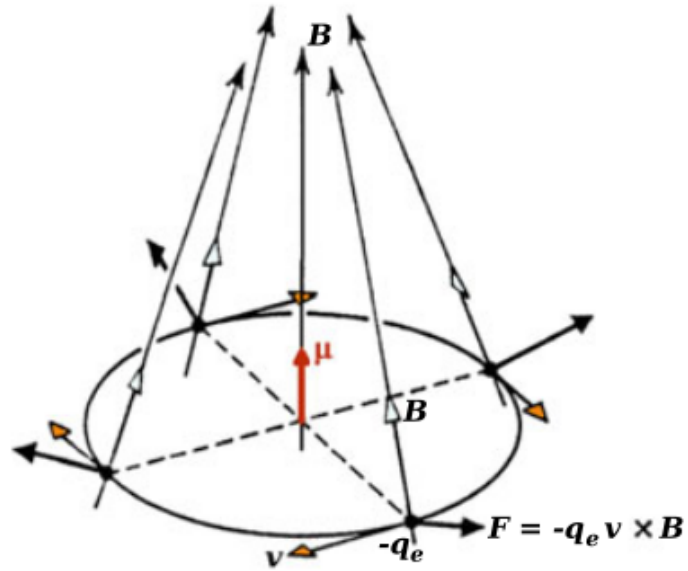


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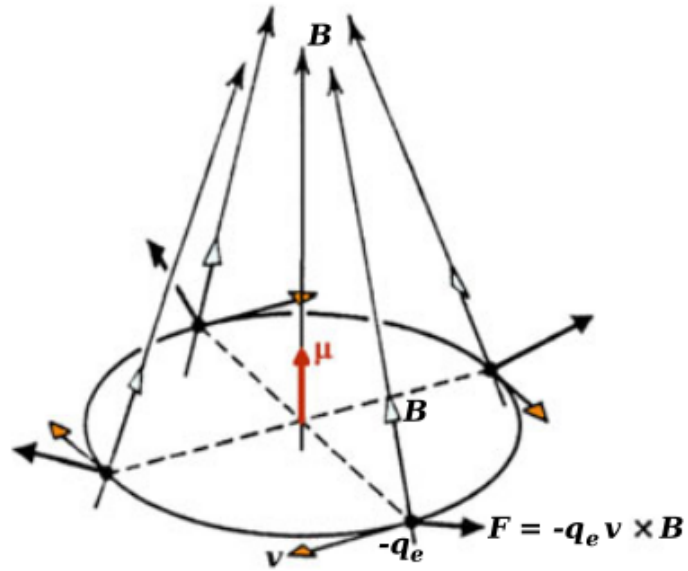


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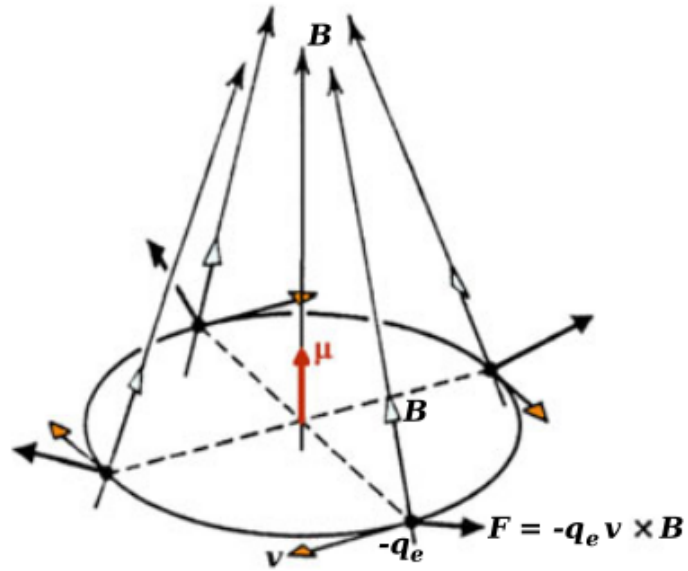


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$$z(t) = z_0 + \dot{z}_0 t + \frac{\mu_z}{2m} \frac{\partial B_z}{\partial z} t^2$$

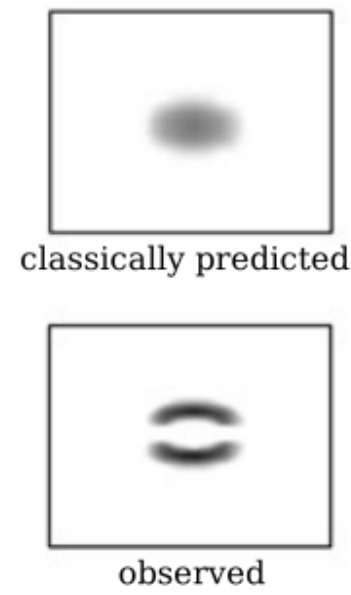
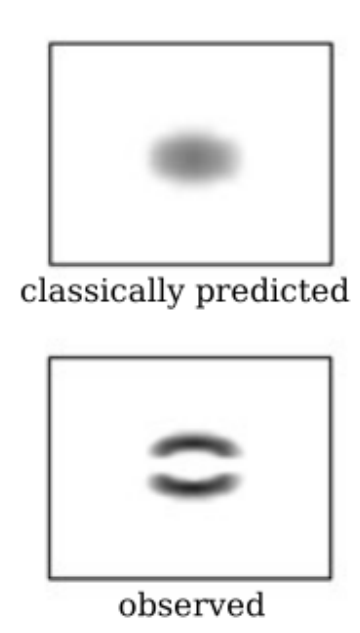


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bla bla bla

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Quantum Mechanics

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- **Normalization**

$$\langle \psi | \psi \rangle = \sum_i (c_i^\psi)^* c_i^\psi = \sum_i |c_i^\psi|^2 = 1$$

$|c_i^\psi|^2$ is the **probability** to find $|\psi\rangle$ in $|e_i\rangle$

A **linear operator** $\hat{O}$ is a transformation from an Hilbert space into itself, that acts like

$$\hat{O}(\alpha|\psi\rangle + \beta|\varphi\rangle) = \alpha\hat{O}|\psi\rangle + \beta\hat{O}|\varphi\rangle = \alpha|\psi'\rangle + \beta|\varphi'\rangle$$

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We can also define the **adjoint** operator $\hat{O}^\dagger$: $\langle\psi'| = \langle\psi|\hat{O}^\dagger$ $\hat{O}_{ij}^\dagger = (\hat{O}_{ji})^*$

We can associate every **observable** of a system to an operator

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Such operators are called **hermitian** and satisfy $\hat{O}_{ij}^\dagger = \lambda_i^* \delta_{ij} = \lambda_i \delta_{ij} = \hat{O}_{ij}$

We define the **commutator** between two operators $\hat{A}$ and $\hat{B}$ as

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Two observables can be **simultaneously** determined iff their operators are compatible, that is if they share a **common basis** of eigenstates:

$$\hat{A}|e_i\rangle = \lambda_i|e_i\rangle \quad \hat{B}|e_i\rangle = \mu_i|e_i\rangle \quad \Leftrightarrow \quad \langle e_i|[\hat{A}, \hat{B}]|e_j\rangle = (\lambda_i\mu_i - \lambda_j\mu_j)\langle e_i|e_j\rangle = 0$$

Position and impulse operators
don't commute!

$$[\hat{p}, \hat{x}] = -i\hbar$$

Definition

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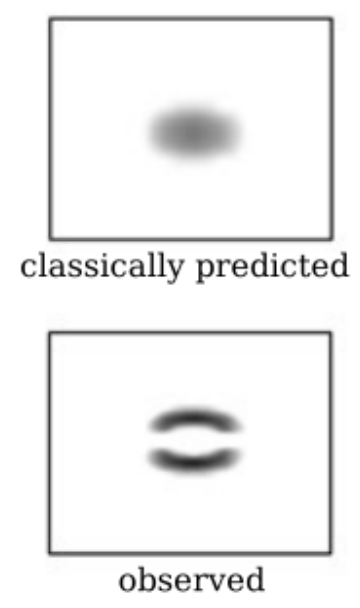
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$$d_l = 2l + 1$$

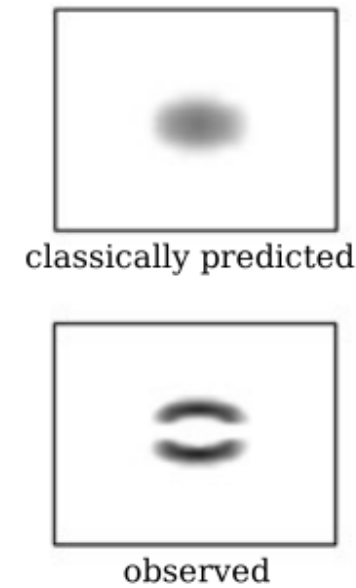


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Figure 14: Stern-Gerlach theoretical versus experimental results.

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Experimental Outcome



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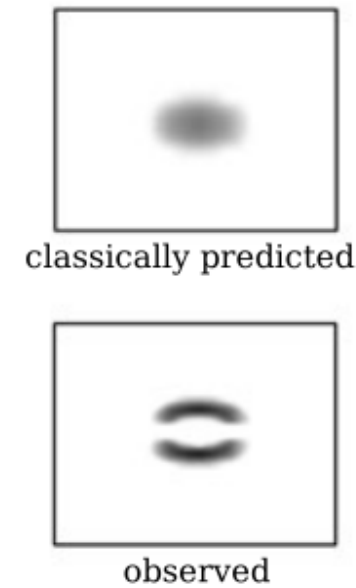
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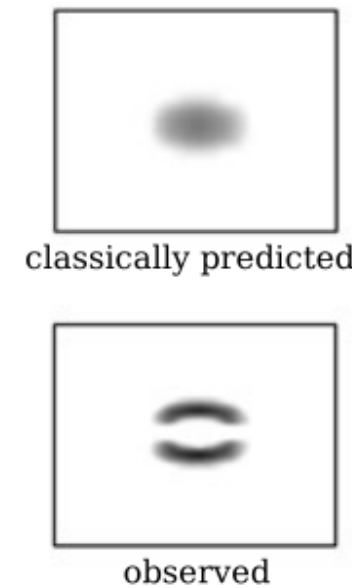


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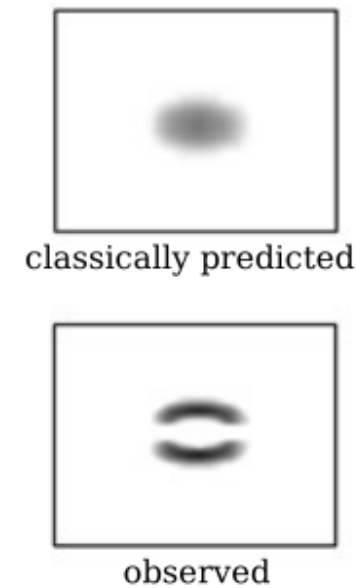


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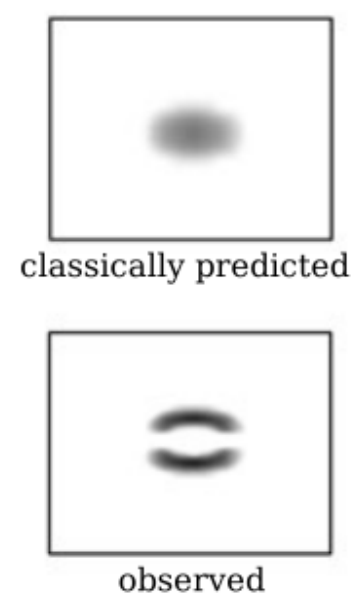
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Degeneracy

The number of states with the same s but different m_s , is:

$$d_s = 2s + 1$$

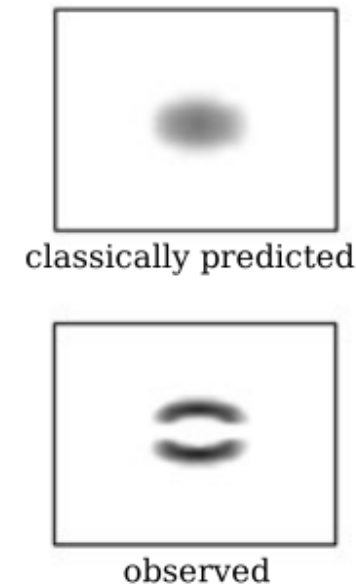


Why do we see two deflected lumps?
Spin!

Total Magnetic Momentum

$$\hat{\mu} = \hat{\mu}_l + \hat{\mu}_s = -\frac{\mu_B}{\hbar} (g_l \hat{L} + g_s \hat{S})$$

Figure 15: Stern-Gerlach theoretical versus experimental results.



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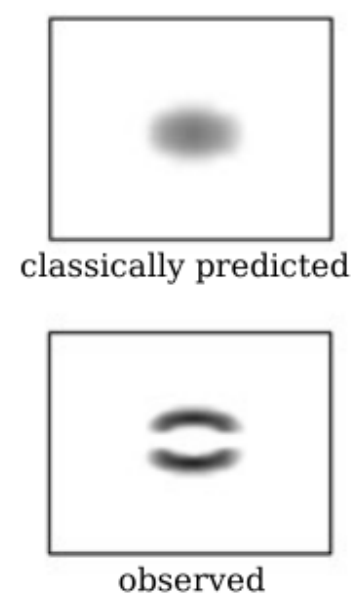
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Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

Spin

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Degeneracy

$$d_j = 2j+1 = (2l+1)(2s+1)$$

Spin

– Total Angular Momentum

Coupling of Angular Momenta

Consider the magnetic field experienced by an e^- orbiting around a nucleus with Z p^+

$$\mathbf{B}(\mathbf{r}) = \frac{Zq_e}{4\pi\epsilon_0 c^2 m_e} \frac{\hat{\mathbf{L}}}{|\mathbf{r}|^3}$$

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We can represent the e^- state using the basis

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Another option is the **coupled** basis

$$\{\hat{\mathbf{J}}^2, \hat{J}_z, \hat{\mathbf{L}}^2, \hat{\mathbf{S}}^2\}$$

since

$$\begin{aligned} [\hat{J}_z, \hat{\mathbf{L}}^2] &= [\hat{J}_z, \hat{\mathbf{S}}^2] = 0 \\ [\hat{\mathbf{J}}^2, \hat{\mathbf{L}}^2] &= [\hat{\mathbf{J}}^2, \hat{\mathbf{S}}^2] = 0 \end{aligned}$$

Energy Spectra in Magnetic Field

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{s-o}} + \frac{\mu_B B_z}{\hbar} (\hat{L}_z + 2\hat{S}_z)$$

— Energy Spectra in Magnetic Field

Energy Spectra in Magnetic Field

Paschen-Back Effect $\mu_B B \gg \xi$

$$\begin{aligned}\xi_{m_l m_s} &= \frac{\mu_B B_z}{\hbar} \left(\langle \hat{L}_z \rangle_\psi + 2 \langle \hat{S}_z \rangle_\psi \right) \\ &= \frac{\mu_B B_z}{\hbar} (m_l + 2m_s)\end{aligned}$$

Spin

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Features

- Spin-orbit coupling negligible
- L and S precess independently around B
- Good quantum numbers: m_l, m_s
- Linear splitting in B

Spin

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Landé g-factor

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

Spin

— Energy Spectra in Magnetic Field

Hello world

Hello world